Entity-Relationship (ER) Model

Entity

- Thing in the real world

Attribute

- Property of an entity
- Most of what we store in the database

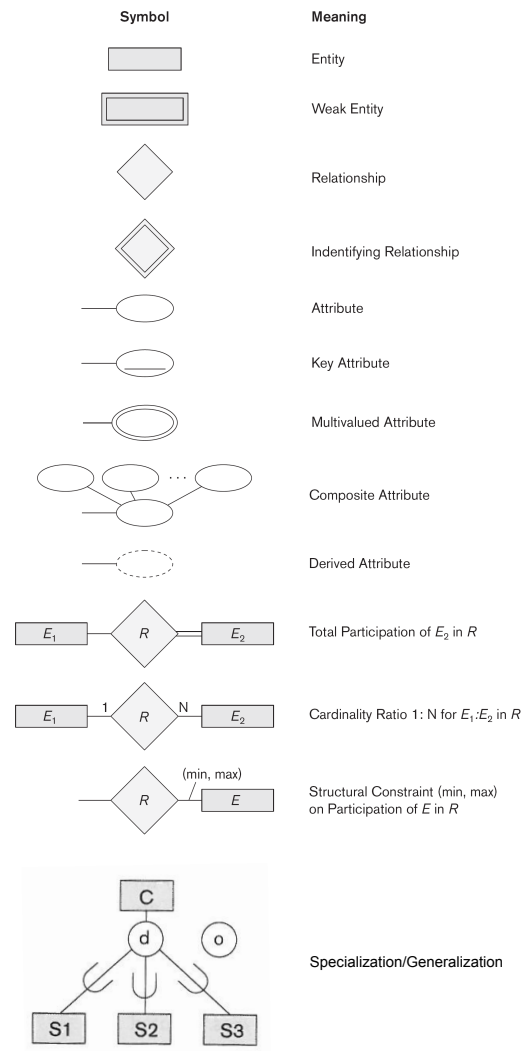
Relationship

- Association between sets of entities
- Possibly with attribute(s)

ER Diagrams

- Graphical depiction of an ER model
- Many notations, this class…

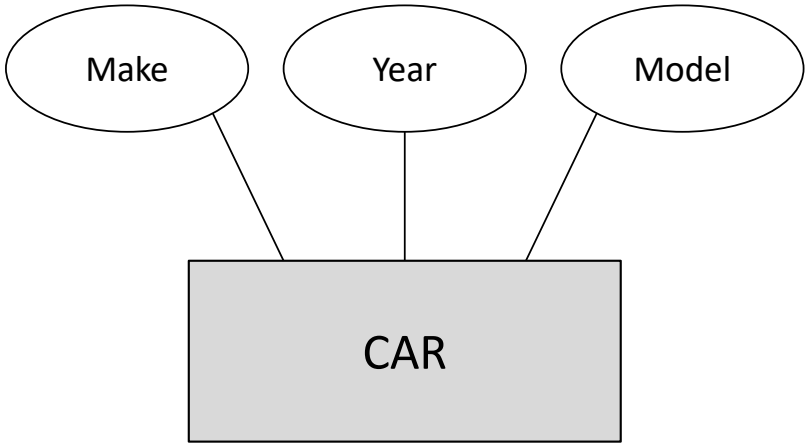
All cars have a year, age, make, model, registration (unique), vehicle number (vin; unique), some number of colors…



Entity Sets

Set of entities that have the same attributes

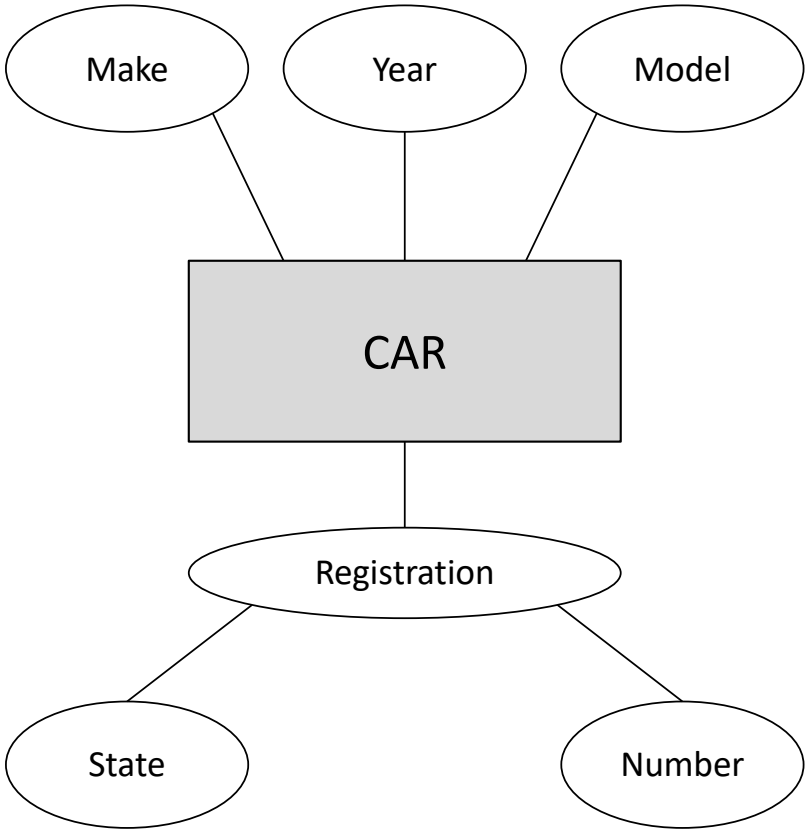
All cars have a year, make, and model.



Composite Attributes

Can be subdivided into smaller subparts

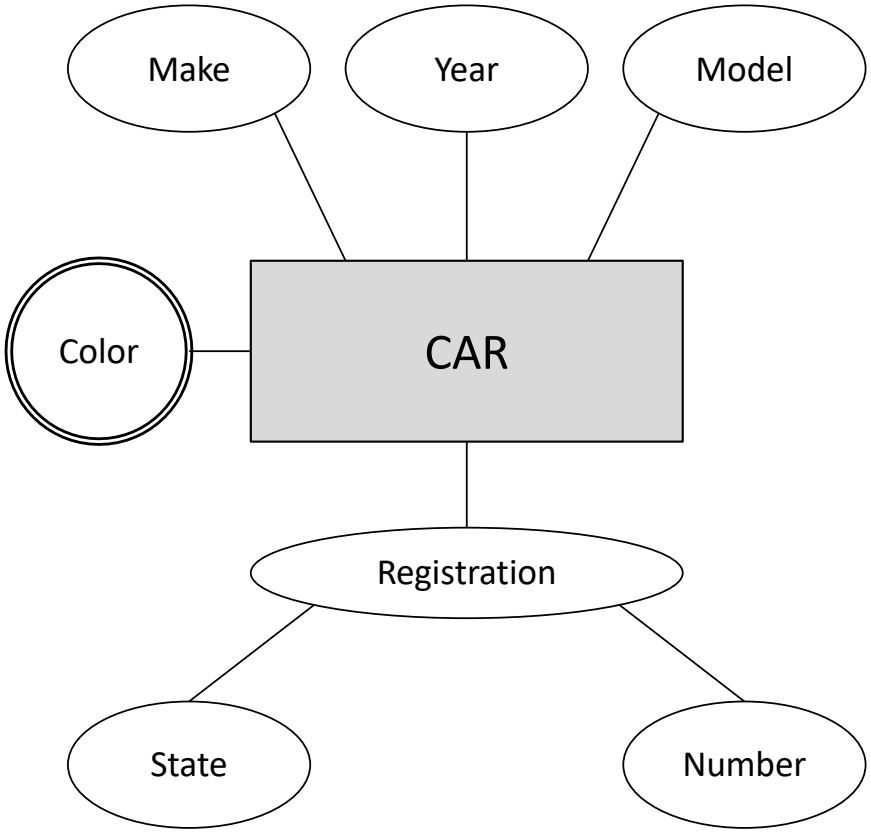
All cars have a year, make, model, ***and registration****.*



Multivalued Attributes

Can take a [possibly specified] number of values.

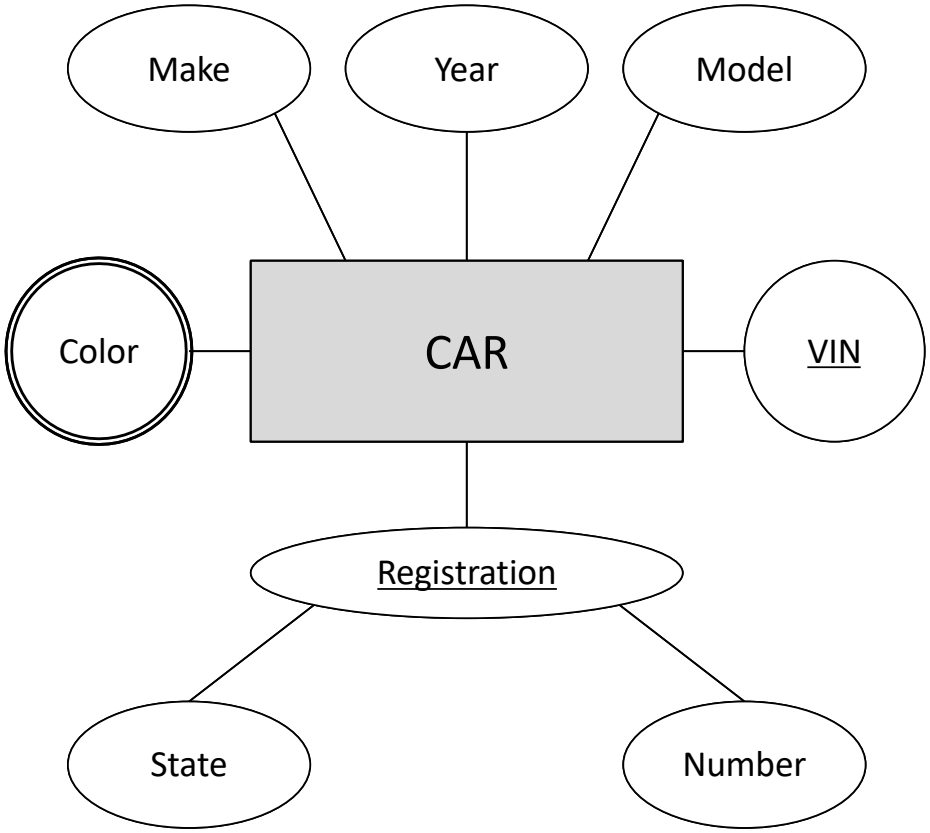
All cars have a year, make, model, registration, and ***some number of colors****.*



Key Attributes

The value uniquely identifies each entity

All cars have a year, make, model, ***registration (unique), vehicle number (vin; unique)****, some number of colors.*



Potential Pitfall

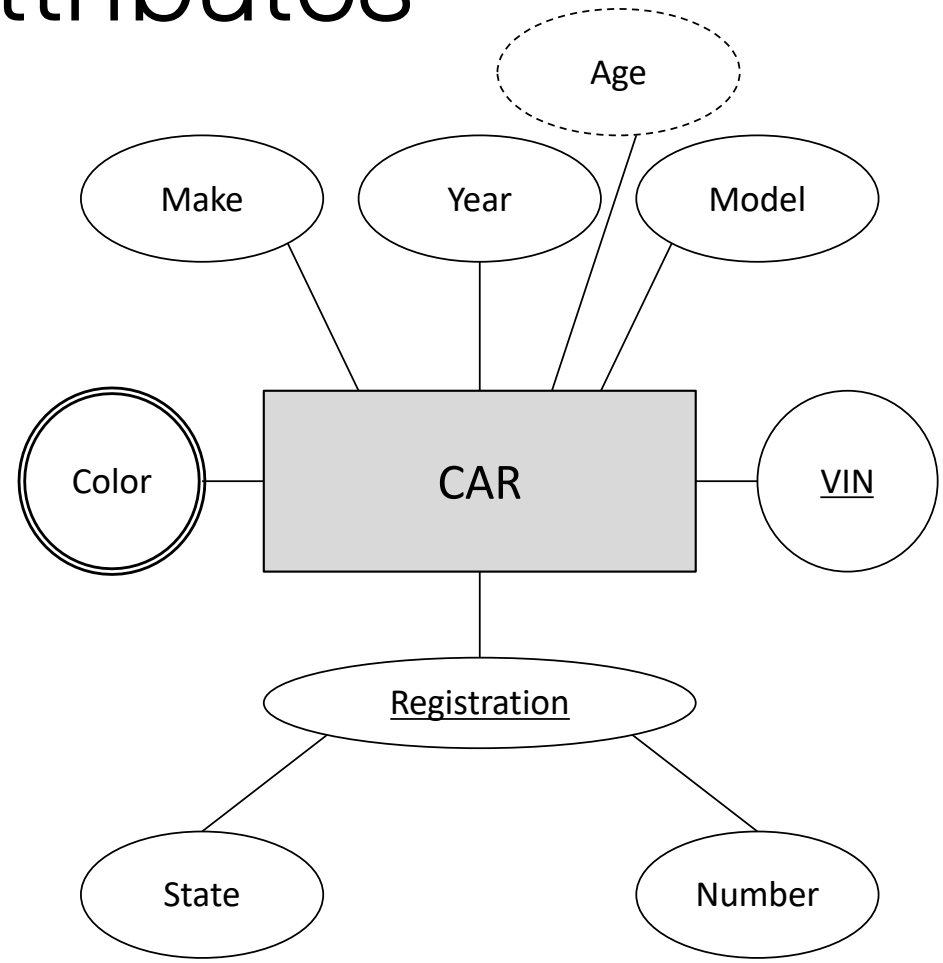
- In relational schema, underlining multiple attributes indicates that for all rows, the *combination* is unique

- In ERDs, underlining multiple attributes indicates that *each individually* can uniquely identify an entity

Derived Attributes

The value can be computed

*All cars have a year, **age**, make, model, registration (unique), vehicle number (vin; unique), some number of colors.*

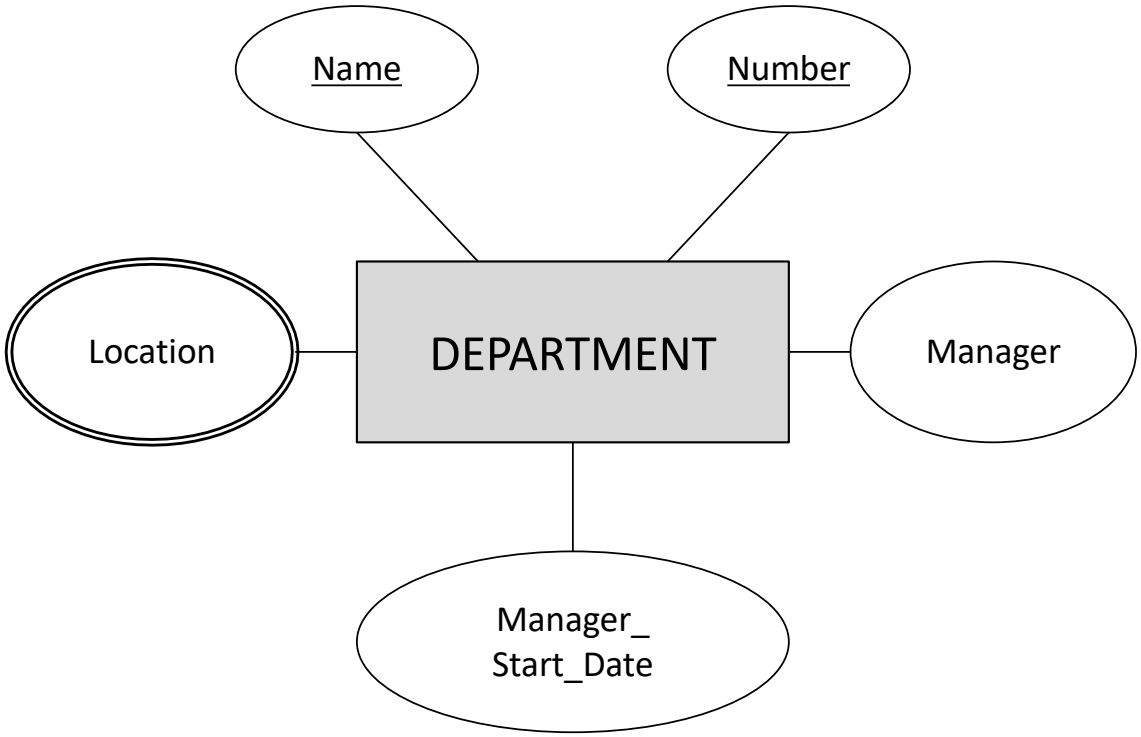


Exercise

Draw an ERD for the following description:

Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.

Answer

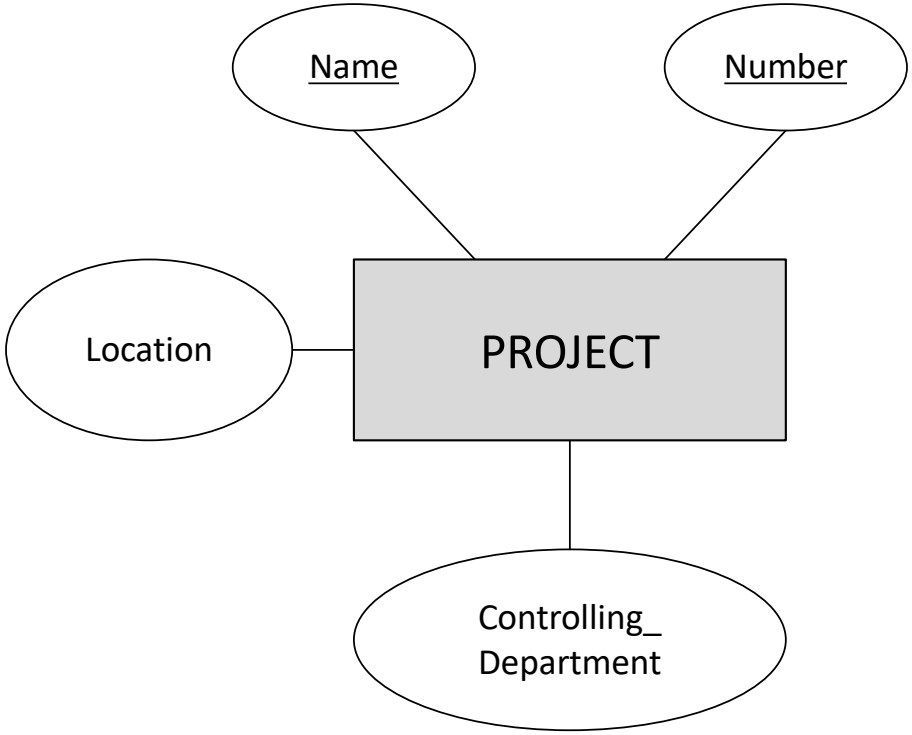

Name
Number
Location
DEPARTMENT
Manager
Manager_
Start_Date

Exercise

Draw an ERD for the following description:

A department controls a number of projects, each of which has a unique name, a unique number, and a single location.

Answer

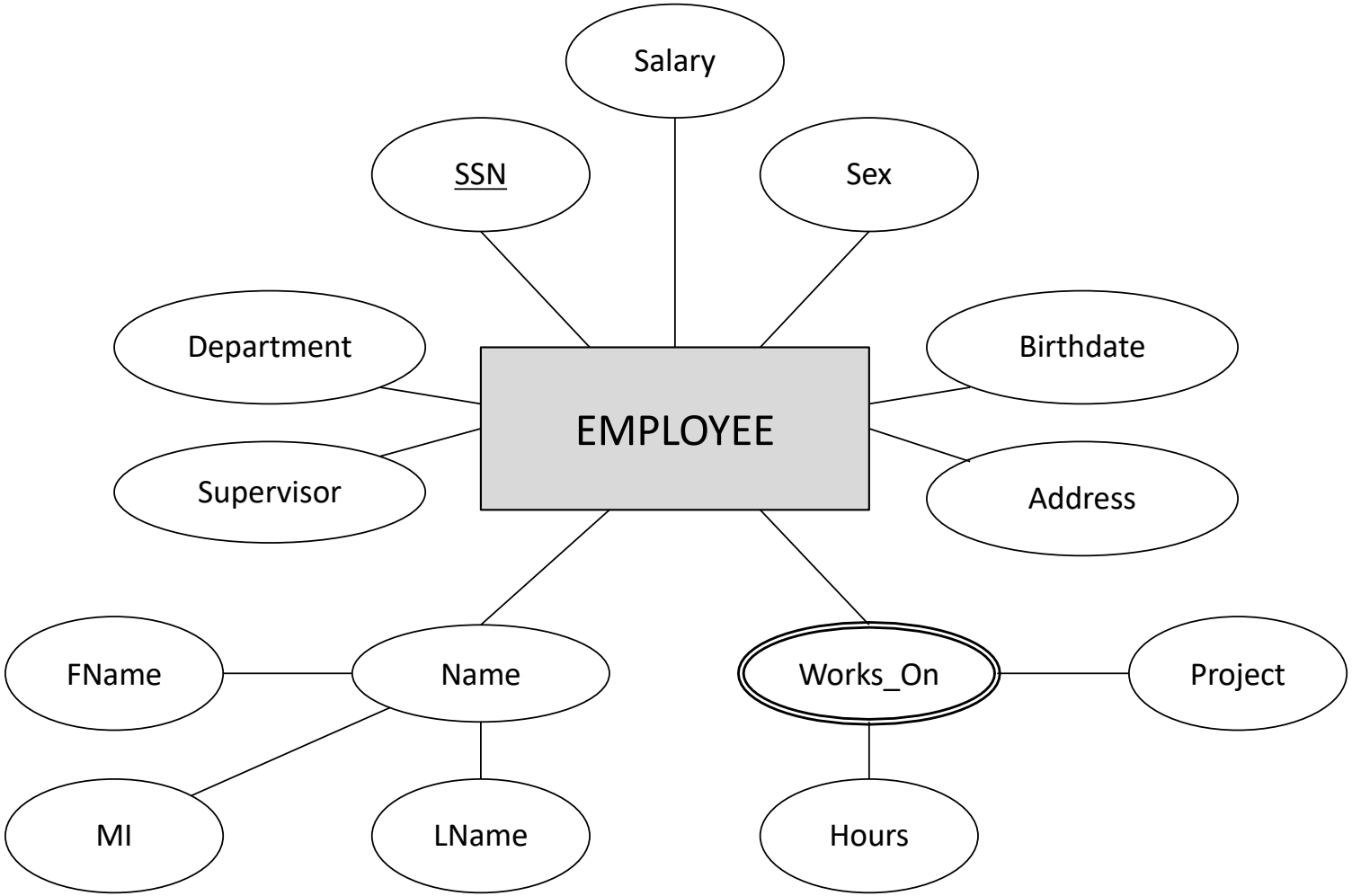


Exercise

Draw an ERD for the following description:

We store each employee's name (first, last, MI), Social Security number (SSN), street address, salary, sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).

Answer

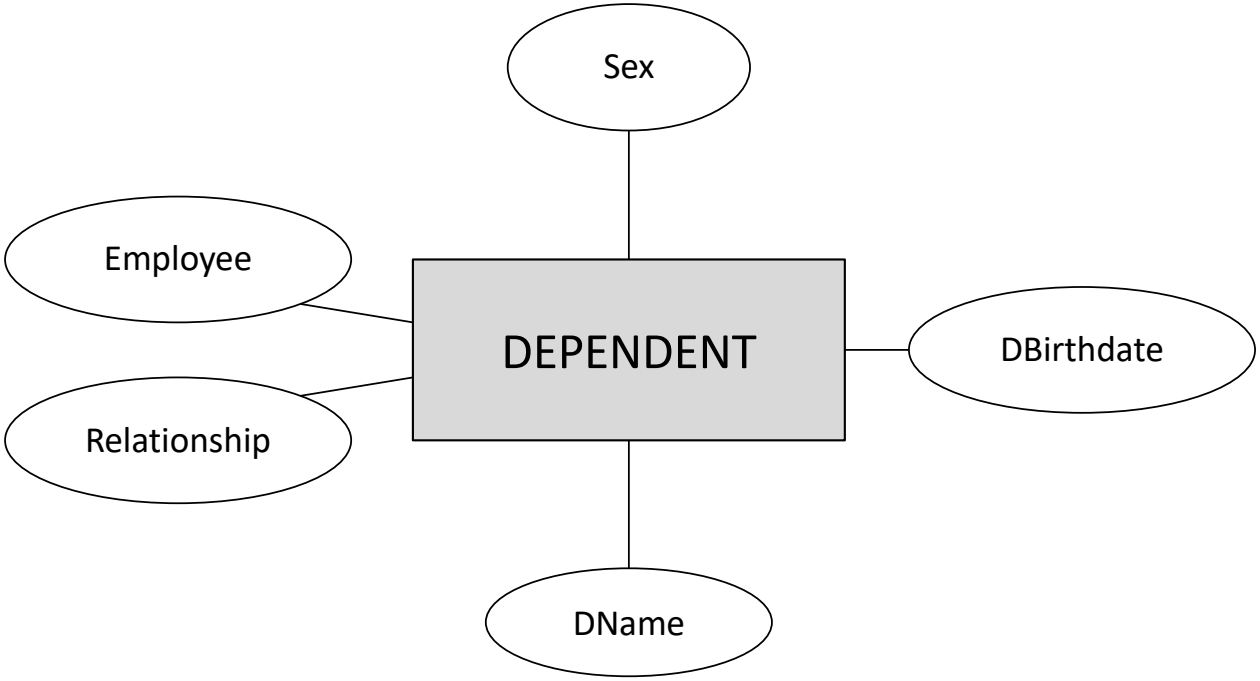

Salary
SSN
Sex
Department
Birthdate
EMPLOYEE
Supervisor
Address
FName
Name
Works_On
Project
MI
LName
Hours

Exercise

Draw an ERD for the following description:

We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, sex, birth date, and relationship to the employee.

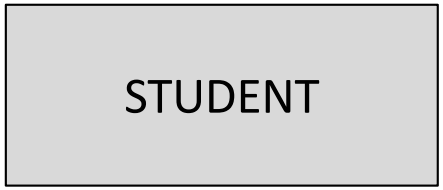
Answer

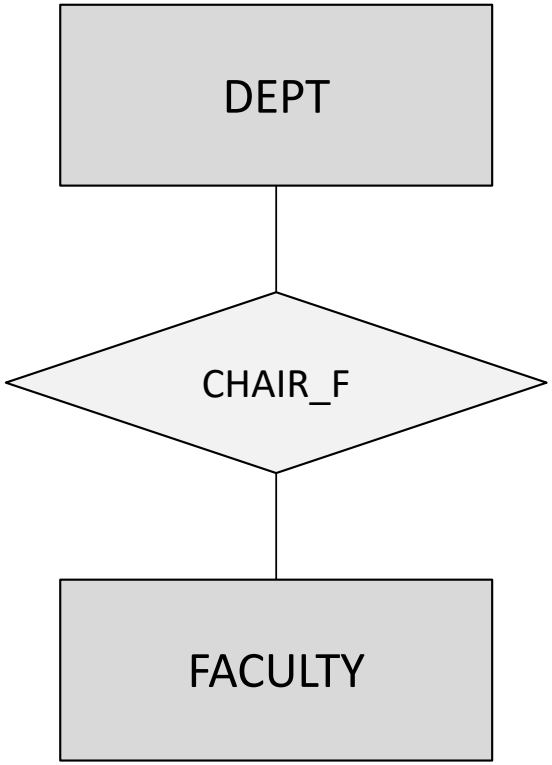


Relationships

Associates one or more sets of entities

– One = recursive (**role** is important)



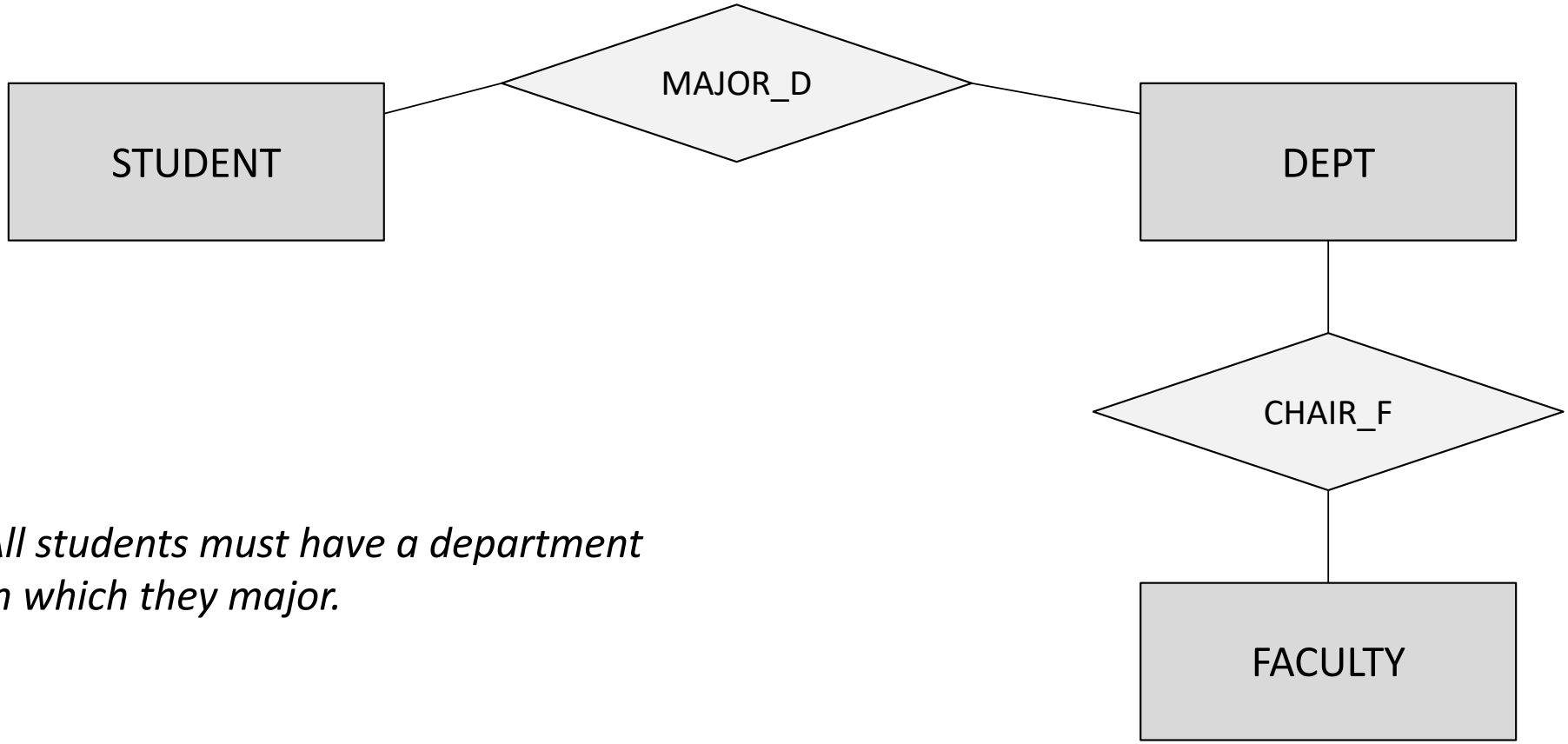


All departments have a faculty member who serves as the chair. A faculty member can only chair one department.

Relationships

Associates one or more sets of entities

– One = recursive (**role** is important)

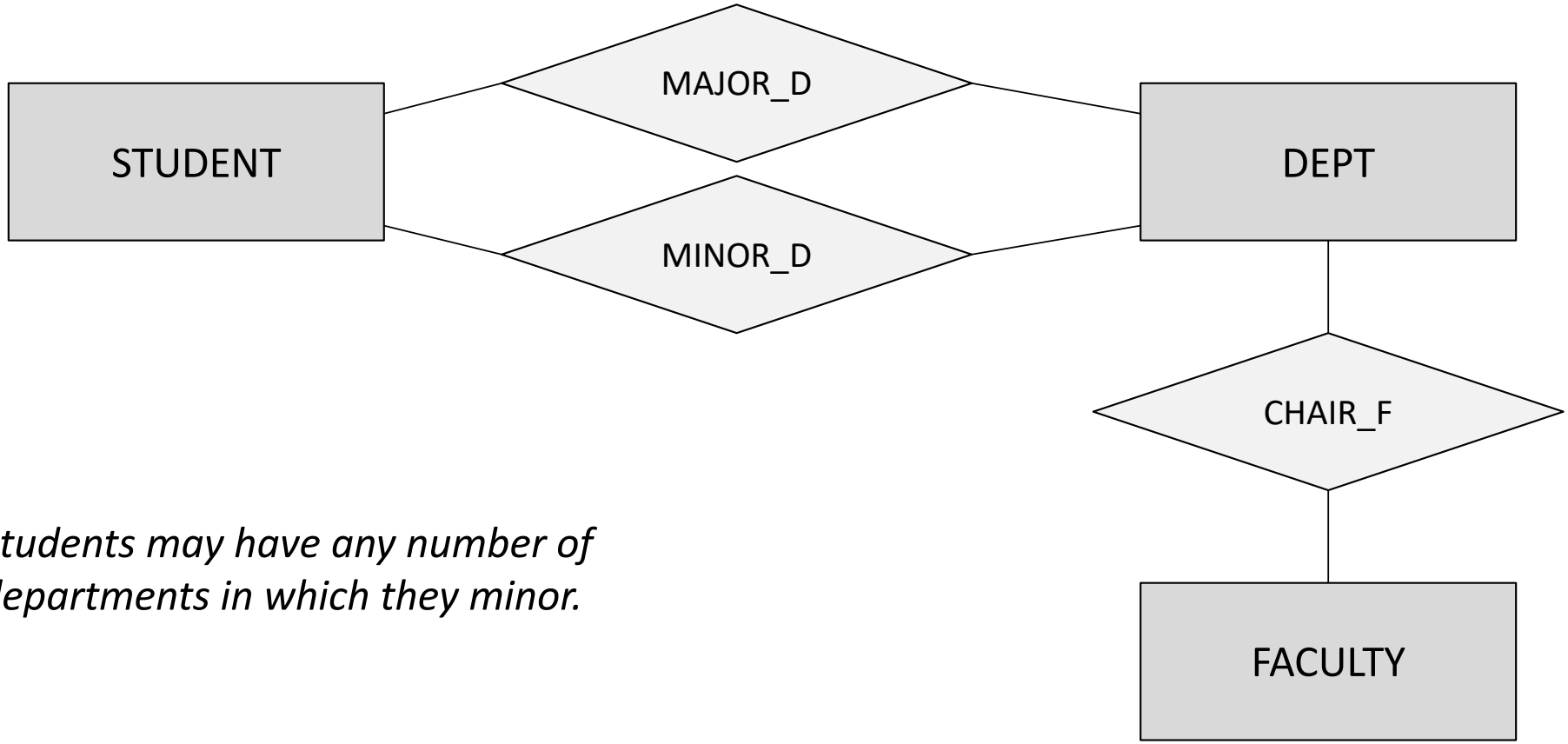


All students must have a department in which they major.

Relationships

Associates one or more sets of entities

– One = recursive (**role** is important)



Students may have any number of departments in which they minor.